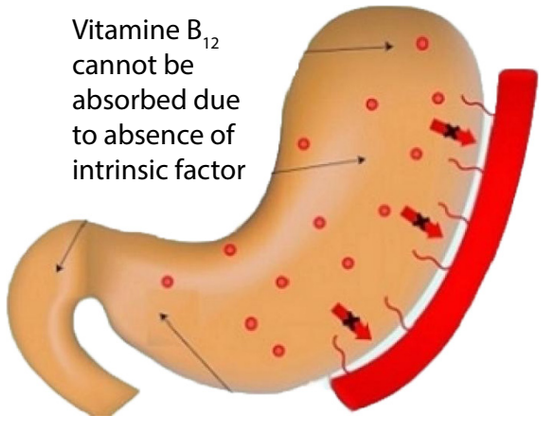
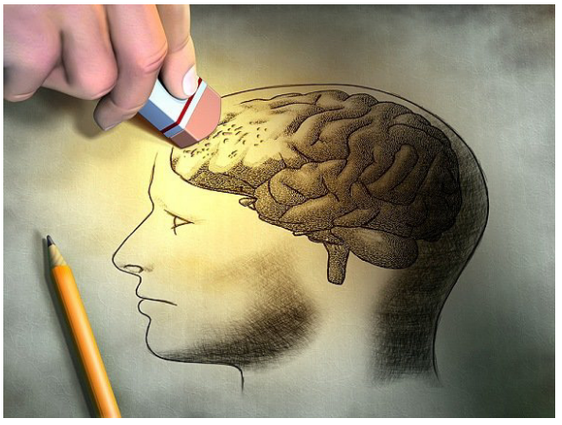
Vitamin B_{12} Deficiency



Pernicious Anaemia



Loss of memory

Fig 11.16: Symptoms of B_{12} deficiency

Functions

- It is essential for the production and regeneration of red blood cells.
- It improves concentration, memory and balance.
- It synthesises and regulates DNA.
- It plays an important role in normal functioning of brain and nervous system.
- It also helps to absorb folic acid.

Food Sources of Cyanocobalamin

Cyanocobalamin is synthesized by bacteria and is found in foods of animal origin. Liver is the richest source of cyanocobalamin. Meat, chicken, oysters, eggs, fish, milk, curd, cheese are good sources of Vitamin B_{12}.

Symptoms of Cyanocobalamin deficiency:

- Loss of memory
- Fatigue
- Anemia
- Severe and irreversible damage to the nervous system and brain
- Pernicious anaemia is caused due to its deficiency in the body which is an immune system disease.

DO YOU KNOW...?

Vitamin B_{12} is unique among all essential nutrients in having a highly specialised mechanism for its absorption. Its absorption from the intestines requires a factor called 'intrinsic factor (IF)' secreted by the stomach. It is a specific protein secreted by the mucosal cells lining the stomach. IF binds and absorbs vitamin B_{12}.

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